Supplementary Material for: Closing Evidence-Blindness in Radiology Report Generators via Dual-Adversarial Causal-Faithfulness Training

Anonymous Authors[1]

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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Table 1. Four-cell decomposition of which filter branch flags each training row. On the 20,560-row training pool, the image-only branch catches an additional 6.5% of rows that the text-only branch alone would miss; this is the symmetric-extension contribution axis relative to ?.

Filter outcome	Count	% of pool
Both filters caught (unimodally solvable on either axis)	15,375	74.8%
Text-only caught, image-only passed	2,156	10.5%
Image-only caught, text-only passed (novelty axis)	**1,329**	**6.5%**
Neither caught (genuinely multimodal)	1,700	8.3%

A. Reproducibility Checklist

Met: (i) Training data sources are public (OpenI; ChestX-Det10 silver-target for cross-site eval; PadChest-GR is named as deferred cross-site target; MIMIC-CXR is not used). (ii) Hyperparameters in Appendix B.4; per-configuration YAML in the released codebase. (iii) 30 unit tests pass on CPU. (iv) Modal H100 80 GB infrastructure documented; all entrypoints documented. (v) Per-fine-tune training-step counts, save-every-steps schedule, and resume-from-checkpoint logic in the released code.

Deferred to extended version: code-release URL, weights-release URL (anonymized for double-blind review; filled at camera-ready); cross-site eval on PadChest-GR (no silver targets available in present infrastructure); three-seed averaging on noise/crop/occlude cross- counterfactuals (single-seed in the main paper's Results section, three-seed values in App. D); mech-interp sweep validating a gradient- norm monitor's AUROC.

B. Extended Methods

B.1. Faith loss derivation and implementation

The naive consistency loss $\mathcal{L}_{\mathrm{KL}} = \mathrm{ReLU}(m - \mathrm{KL}(p_{\mathrm{full}} \| p_{\mathrm{alt}}))$ admits a degenerate optimum: a model that always emits a uniform distribution under masking satisfies any finite KL margin without using the image. The token-wise correctness-gain margin replaces the distributional divergence with a per-token signal coupled to gold-token correctness on the full input. The implementation runs three forward passes per training step (full, image-zeroed, image-flipped on laterality rows) and exposes a flipped-row mask that zeroes the flipped contribution on non-laterality rows. Image-zeroing uses the per-channel mean of the batch's pixel values, not a literal zero tensor: a literal zero is, after the processor's normalisation, a specific out-of- distribution input the model has never seen at training time. The per-channel mean produces a uniform "neutral" image at the model's own normalisation, which is what we want when we say "the image is absent."

B.2. Dual filter four-cell decomposition

After the subsequent text-target filter restricts training to OpenI rows with non-empty findings text, the per-row weight distribution within the surviving 10,839 rows is 95.6% at $w_{\min} = 0.2$ and 4.4% at $w = 1.0$. The continuous-SFC ablation replaces the discrete $\{w_{\min}, 1\}$ rule with $\mathrm{SFC}(x) = 1 - p_{\mathrm{text}}^{\mathrm{correct}}(x)$ clipped to $[w_{\min}, 1]$.

B.3. IMG_correct derivation and the "IMG_correct > IMG" identity

For a binary classifier f with gold class y, partition examples into cc/ci/ic/ii by whether f is correct on the full and image-zeroed inputs. Then $\mathrm{IMG} = (n_{\mathrm{ci}} - n_{\mathrm{ic}})/n \times 100$. IMG_correct further partitions ci into a margin-passing subset (where $p_{\mathrm{full}}(y) - p_{\mathrm{masked}}(y) \geq \tau$) and a margin-failing subset (the induced-inversion sub-population):

$$\begin{aligned} \mathrm{IMG\_correct} &= n_{\mathrm{ci,with\text{-}margin}}/n \times 100, \\ \text{induced inv.} &= (n_{\mathrm{ci,no\text{-}margin}} - n_{\mathrm{ci,with\text{-}margin}})/n \times 100. \end{aligned}$$

IMG_correct can exceed IMG. When the ic population (model becomes correct under masking) is non-trivial and the margin- failing ci population is small, IMG_correct $> (n_{\mathrm{ci}} - n_{\mathrm{ic}})/n$. This is why several configurations in the retroactive audit show IMG_correct > IMG; it is mathematically valid and does not indicate negative induced inversion. For autoregressive generators we apply this logic per token on the gold target sequence (with

Table 2. Training hyperparameters across all five configurations. The configurations differ only in (i) which weights JSONL is loaded, (ii) whether the faith loss is enabled, (iii) whether the flipped counterfactual is enabled, and (iv) the resulting λ_{faith}. All other hyperparameters are identical.

Hyperparameter	Value
Backbone	MedGemma-4B-IT (research-only licence)
LoRA rank / alpha	16 / 32
LoRA targets	q_proj, k_proj, v_proj, o_proj
Trainable parameters	11.9 M (0.28% of 4.31 B base)
Optimizer	AdamW
Learning rate	2×10^{-4}
Weight decay	0.0 on LoRA targets, 0.01 elsewhere
LR schedule	cosine to zero, 50-step linear warm-up
Batch size	2 (grad accum $\times 8$; effective 16)
Max optimizer steps	2,500
Mixed precision	bf16
Faith-loss margin (nats)	0.3
Image masking	per-channel mean of batch pixel values
λ_{faith}	0.3 when enabled, 0 otherwise
$w_{\min}$ (filter downweight)	0.2
Filter activation threshold	0.7 on $p_{\text{branch}}^{\text{correct}}$
Random seeds	$\{42, 1337, 9001\}$
GPU	NVIDIA H100 80 GB (Modal)
Wallclock per fine-tune	1.5–4.0 h (faith-loss configs $\sim 3\times$ slower)

the causal-LM shift between logits and target IDs).

B.4. Training hyperparameters and compute

Faith-loss configurations are $\sim 3\times$ slower per step than simple-loss configurations because each step runs three forward passes (full, image-zeroed, image-flipped). Total Phase 4 training compute: $\approx$ \$200 on Modal H100 80 GB across 15 fine-tunes (5 configs $\times$ 3 seeds). Total project envelope (training, primary evals, retroactive audit, cross-counterfactual, cross-site): $\approx$ \$400.

C. Full IMG_correct Retroactive Audit

Distribution of induced-inversion contributions. Across all 18 cells: induced inversion has an absolute range of $[0.00, 1.31]$ pp and a relative range of $[0.0\%, 13.2\%]$ of the corresponding IMG. The cell with the largest relative induced inversion is v5_0_base at 13.2%, with the smallest absolute IMG (1.28 pp); fractionally large because any flipping at all under masking is fractionally large at this scale. The cell with the largest absolute induced inversion is v6_0_loo_no_padchest at 1.31 pp (3.1% of IMG). No cell exceeds the pre-registered $\geq 30\%$ threshold.

The abl_no_consist cell is the cleanest disconfirmation of the pre-registered concern. Removing the consistency loss collapses IMG_correct from ~ 70 pp (nominal v5 family) to 3.60 pp — near baseline — while simultaneously reducing induced inversion from ~ 0.7 pp to 0.30 pp. The pre-registered concern was that consistency-loss training caused induced inversion. The empirical pattern is the opposite: with the consistency loss, IMG_correct is high and induced inversion is small; without it, both are small.

D. Cross-Counterfactual Detail

Baseline anchor. On the matching crop counterfactual, sft_only seed-42 yields IMG = 0.33 pp, IMG_correct = 2.80 pp at 73.82% accuracy, induced inversion = 0.26 pp (n = 105,921 tokens). The cross-counterfactual sft_full − sft_only IMG_correct gap on crop is $4.81 - 2.80 = 2.01$ pp; the gap on the trained-mean counterfactual is $74.00 - 2.38 = 71.62$ pp. The order-of-magnitude difference between trained-mean and crop is the operational meaning of the partial- circularity finding.

Table 3. Full IMG_correct retroactive audit, all 18 cells (nine verifier configurations + nine v5 ablations, each on $n = 2{,}974$ verifier rows). Acc full and Acc masked are accuracies under the full and image-zeroed inputs, respectively.

Config	IMG_corr	IMG	Inv	Inv/IMG	Acc full	Acc mask
Headline configurations.						
v5_0_base	3.83	1.28	0.17	13.2%	92.30	91.02
v5_1_ground	3.40	2.62	0.13	5.1%	91.22	88.60
v5_2_real	64.66	61.84	0.91	1.5%	92.60	30.77
v5_3_contrast	71.05	68.12	0.74	1.1%	93.01	24.88
v5_4_final	68.43	63.55	0.40	0.6%	91.69	28.14
v6_0_3site	74.51	70.21	0.07	0.1%	91.06	20.85
v6_0_loo_no_openi	38.20	34.20	0.00	0.0%	87.42	53.23
v6_0_loo_no_padchest	43.95	41.80	1.31	3.1%	92.30	50.50
v6_0_loo_no_chestx	18.59	14.53	0.20	1.4%	81.27	66.75
v5 ablations.						
abl_scale_25	69.07	65.64	0.71	1.1%	92.70	27.07
abl_scale_50	70.51	67.59	0.81	1.2%	92.80	25.22
abl_scale_100	66.24	62.81	0.61	1.0%	92.80	29.99
abl_hothresh_70	73.84	69.97	0.57	0.8%	92.60	22.63
abl_hothresh_80	77.98	74.78	0.74	1.0%	92.77	17.99
abl_no_contrast	63.28	60.02	0.87	1.4%	92.77	32.75
abl_no_ground	72.53	69.64	0.50	0.7%	92.97	23.34
abl_no_uncert	73.40	70.54	0.98	1.4%	93.17	22.63
abl_no_consist	3.60	2.79	0.30	10.8%	88.80	86.01

Table 4. Cross-counterfactual eval — full numbers including masked accuracy. Each row uses the same sft_full seed-42 checkpoint; only the counterfactual transformation differs.

Counterfactual	IMG	IMG_corr	Acc full	Acc mask
mean (TRAINED)	73.97	74.00	75.48	1.51
noise	62.10	62.20	75.48	13.38
crop	3.64	4.81	75.48	71.83
occlude	0.28	1.11	75.48	75.20

E. Cross-Site Detail

Silver target construction. ChestX-Det10 test rows in the standard test set have empty evidence_text fields. We constructed groundbench_test_chestx_det10_silver.jsonl (1,122 rows) by injecting CheXagent-2-3B (?) generations as reference_report; the silver-target nature is reported explicitly. PadChest-GR cross-site (?) requires a fresh CheXagent generation pass and is deferred. Token-level analysis. The cross-site 54.65% token-level accuracy is 20 pp lower than the in-distribution OpenI accuracy, reflecting (i) genuine cross-site difficulty (different camera vendors, different radiologist phrasing) and (ii) the silver- target nature (the model is being graded against another model's output, not radiologist gold).

F. Pre-Registered Hypotheses

(i) Full method reduces IMG_correct on OpenI-test by ≥ 5 pp vs. SFT-only at matched accuracy. Status: HIT BY AN ORDER OF MAGNITUDE on three seeds $\times$ three faith-loss configurations. Realised: +72.58 pp at +1.17 pp accuracy.

(ii) IMG_correct refinement reveals $\geq 30\%$ of v5_3_contrast's IMG as induced inversion. Status: REFUTED. Realised: 1.1%. Pattern holds across all 18 audited cells (induced-inversion contributions ≤ 1.31 pp absolute, $\leq 13.2\%$ relative).

(iii) Dual filter dominates the faith loss in the ablation. Status: INVERTED. The empirical pattern is the opposite: sft_dual is statistically indistinguishable from sft_only; the faith loss is the load-bearing component (sft_faith alone gives +71.73 pp over baseline). The increment from adding the dual filter on top of the faith loss is +0.85 pp seed-mean (within one standard deviation of sft_full). We retain the dual filter in the recipe because the four-cell decomposition (App. B.2) demonstrates a symmetric-extension contribution axis not addressable by text-only filtering.

One pre-registration HIT, one REFUTED, one INVERTED — all reported transparently. The methodological refinement (IMG_correct) lands as a contribution independently of the empirical results.